

Lab 4

1. Give self-contained definitions of the following languages/sets: A_{TM} (the acceptance problem), $Halt_{TM}$ (the halting problem), TM M decides Language L , TM M semidecides Language L .
2. Show that A_{TM} is recognizable (construct a recognizer for it) .
3. Show that A_{TM} is undecidable. (You are required to use the diagonalization technique)
4. Is $\overline{A_{TM}}$ recognizable? Explain your answer.
5. Give the definitions of the following classes:
 - 1) Nondeterministic TM M decides a Language L
 - 2) Nondeterministic TM M semidecides a Language L
6. State the Church-Turing Thesis
7. Prove the following language is not recursively enumerable.

$\overline{H_1} = \{w : \text{either } w \text{ is not the encoding of a Turing machine, or it is the encoding "M" of a Turing machine } M \text{ that does not halt on "M"}\}.$